Recitation 1

Problem 1 - Vectors

A small online store records the number of orders received during 12 consecutive hours: 18, 24, 15, 31, 27, 22, 35, 19, 28, 40, 26, 33.

1. Create a vector called `orders` containing these values.
2. Use indexing to extract:

- the first four observations
- the 3rd, 6th, 9th, and 12th observations
- all observations except the first and last

3. Create a new vector called `high_orders` containing only values from `orders` that are greater than 25.
4. The store later discovers that the 4th recorded value should have been 29 rather than 31. Modify `orders` accordingly.
5. Insert the value `21` immediately after the second observation and save the resulting vector as `order_updated`.
6. Using a vectorized operation, create a vector called `adjusted_orders` in which every value of `ordered_updated` is increased by 10%.
7. Create the following vector without typing every element individually: 5, 10, 15, 20, 25, 30, 35, 40.

Problem 2 - Lists

Suppose we want to store information about a statistical computing course. Create a list called `course_info` containing the following five elements: "Applied Statistical Computing", 35, the vector ["R", "Python"], the vector [88, 92, 79, 95, 84], another list containing "Fall 2026" and TRUE.

1. create `course_info`.
2. Display the third element of `course info` using single ([]) and double ([[]]) brackets. Examine the outputs. What is the difference?
3. Extract `Python` directly from `course_info`.
4. Extract the value `95` directly from `course_info`.
5. Extract `Fall 2026` directly from `course_info`.
6. Change the second element of `course_info` from 35 to 40.
7. Change the second programming language from "Python" to "SQL".
8. Add "Statistics Department" as a new sixth element of `course_info`.
9. Remove the fourth element of the list.

10. Create another list called `instructor_info` containing "Dr. Smith" and "Columbia University". Combine `course_info` and `instructor_info` using `c()` and save the result as `combined1`. Next, combine `course_info` and `instructor_info` using `list()` and save the result as `combined2`.Compare `combined1` and `combined2`. How many elements does each contain?

Problem 3 - Matrices

The following values represent quiz scores for 4 students on 3 quizzes:

- 82, 75, 91;
- 68, 84, 79;
- 95, 88, 93;
- 76, 81, 87.

1. Create a matrix called `scores` with 4 rows and 3 columns, arranging the values by row.
2. Extract:

- Student 3's Quiz 2 score;
- all scores for Student 2;
- all Quiz 3 scores;
- the scores for Students 1 and 4 on Quizzes 1 and 3.

3. Student 2's Quiz 1 score was entered incorrectly. Replace 68 with 73.
4. Add a fourth quiz using the scores 90, 85, 96, 82.
5. A fifth student joins the dataset with scores 80, 77, 89. Add this student as a new row to `scores` and call the resulting matrix `scores2`.
6. Create a new matrix called `bonus_scores` by adding 3 points to every score in `scores2`. Do this using a vectorized matrix operation.
7. In `bonus_scores`, replace every score greater than 95 with 100.
8. Extract only those elements of `bonus_scores` that are at least 90.
9. Remove Student 2 and Quiz 1 from `bonus_scores` and save the resulting matrix as `reduced_scores`.

Problem 4 - Data Frames

A company collected information about eight employees:

- Employee - ["Anna", "Ben", "Carla", "David", "Emma", "Frank", "Grace", "Henry"]
- Department - ["Analytics", "Marketing", "Analytics", "Finance", "Marketing", "Analytics", "Finance", "Analytics"]
- Experience - [2, 7, 5, 9, 3, 6, 8, 4]
- Salary - [68000, 85000, 79000, 105000, 72000, 92000, 99000, 76000]
- Remote - [TRUE, FALSE, TRUE, FALSE, TRUE, FALSE, TRUE, TRUE]

1. Use these vectors to create a data frame called `employees`.
2. Use appropriate R commands to examine:

- the structure of `employees`;
- a summary of its variables;
- the first three rows;
- its dimensions.

3. Extract the `Salary` variable in three different ways so that:

- one result is a data frame;
- the other two results are vectors;

4. Extract only the `Employee`, `Department`, and `Salary` columns.
5. Extract rows 2 through 6 and only the `Employee` and `Experience` columns.
6. Extract all employees who have at least 5 years of experience.
7. Add a new variable called `Salary_K` that contains salaries expressed in thousands of dollars. For example, 68000 becomes 68.
8. Add another variable called `Senior` that takes the value `TRUE` when `Experience` is at least 6 years and `FALSE` otherwise.
9. Carla's salary should be 82,000 instead of 79,000. Modify the appropriate element of the data frame.
10. Add the following employee as a new row: Employee - Irene; Department - Marketing; Experience - 10; Salary - 110000; Remote - FALSE.